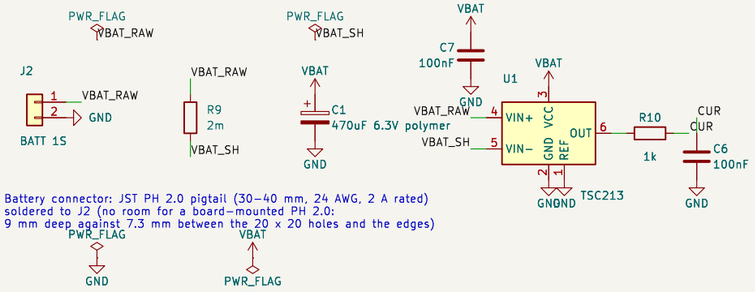
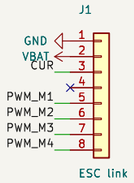
A — Battery entry, current shunt, current sense



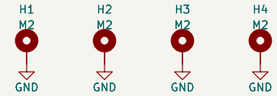
Battery connector: JST PH 2.0 pigtail (30–40 mm, 24 AWG, 2 A rated) soldered to J2 (no room for a board-mounted PH 2.0: 9 mm deep against 7.3 mm between the 20 x 20 holes and the edges)

Battery current: CUR = 0.1 V/A on J1 pin 3 (FC ADC PA4); discharge only. Motor peak about 8 A -> 0.8 V; full scale 31 A at 3.1 V. Shunt loss at 6 A: 72 mW. U1 senses R9 through two Kelvin tracks.

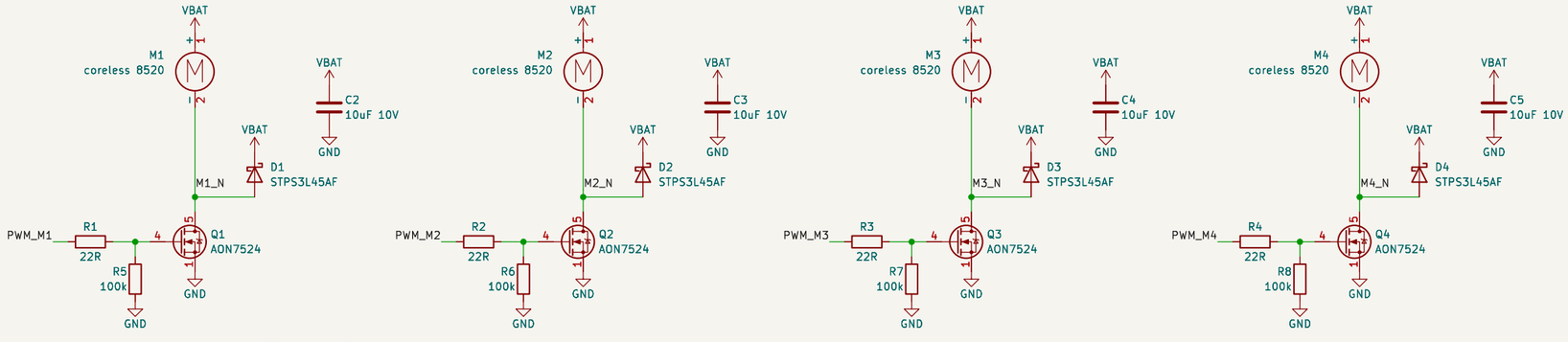
B — Link to the flight controller (standard 4-in-1 ESC connector)



Common 4-in-1 ESC pinout. A brushless 20 x 20 1S 4-in-1 ESC plugs into the same FC cable: check its pinout first (some vendors swap VBAT/GND or CUR/TLM).
TLM (pin 4) is not used by this brushed board.
VBAT on pin 2 also supplies the whole FC (up to 0.5 A at 4.1 V).
VBAT is off while the charge port is plugged in (Q5): FC and motors off.

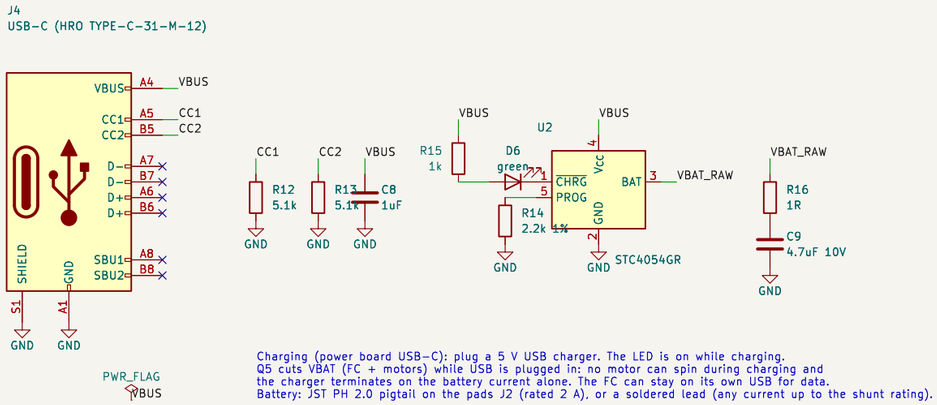


C — Motor stages: 4 x low-side N-MOSFET, 8520 coreless on 1S



Rg 22 ohm limits the FC output peak current (Ciss 2250 pF). Rpd 100k keeps the MOSFETs off while the FC boots or when the cable is unplugged. Brushed PWM 32 kHz from the FC (TIM2 CH1–CH4).

D — USB-C charge port (power only), 1S charger STC4054, charge switch Q5



Charging (power board USB-C): plug a 5 V USB charger. The LED is on while charging.
Q5 cuts VBAT (FC + motors) while USB is plugged in: no motor can spin during charging and the charger terminates on the battery current alone. The FC can stay on its own USB for data.
Battery: JST PH 2.0 pigtail on the pads J2 (rated 2 A), or a soldered lead (any current up to the shunt rating).

Generated automatically: run ERC before release
Larger drones: a commercial 4-in-1 ESC replaces this board on the same FC cable
FC link: standard 4-in-1 ESC connector, JST SH 8-pin, 1:1 cable
40 x 40 mm, 4 layers, 1.0 mm; frame C outline; 20 x 20 mm M2 holes (hard mount)

ACwing — Andrea Cirillo

Sheet: /
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Title: ACwing ESC 1S rev A — 1S brushed power board with USB-C charger,

Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 1/1